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Frequency Response of a Shuttered Open Jet Wind Tunnel

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Steady and unsteady response of an open-jet Eiffel-type wind tunnel to a shuttering system was investigated using hot wire anemometry and flow visualization. Vertical shutters (louvers) were installed in the diffuser behind the test section. Smoke visualization revealed a shear layer with large discrete vortices, bounding the test-section core flow. The scale and frequency of these vortices varies with airspeed and shutter frequency. For a sinusoidal shutter motion, the velocity in the test section varies as a function of $\sin^n(\pi f t)$ with maximum value of $n = 3$. The mean test section flow speed is independent of the shutter actuation frequency, while the RMS correlates with the ratio of the shutter frequency and tunnel RPM. Flow visualization gives a shear layer expansion angle of roughly 20° for all cases. The vortex convective speed in the shear layer is also independent of the shutter frequency.

Nomenclature

C_P	Coefficient of Pressure
C_Q	Discharge Coefficient
U_{Shear}	Mean Shear Layer Velocity (m/s)
f	Shutter Actuation Frequency (Hz)
β	Shutter Angle (Degrees)
A_{Exit}	Test Section Exit Area (m)
U_β	Test Section Instantaneous Velocity (m/s)
$U_{\beta_{Static}}$	Test Section Instantaneous Velocity at fixed β (m/s)
U_δ	Test Section Velocity at RMS(β) (m/s)
t	Time (s)
D_{Fan}	Tunnel Fan Diameter (m)
Ω	Tunnel Fan RPM
U_{Vortex}	Vortex Velocity (m/s)
RMS	Root Mean Square

1. Introduction

Noting lack of comprehensive definition of "gusts" [1], especially those relevant to small aircraft in low-altitude operations, we consider here a mechanism for generating large vortical disturbances in a low-speed ground-test facility. Spatial or temporal length scales of atmospheric gusts can vary from few millimeters to several kilometers [2], with additional complications near the ground [3]. Recent interest in flight-relevant ground test aimed at small/light/slow vehicles [4], and wind-energy applications [5] has engendered vigorous recent interest in unsteady wind tunnels [6]. Unsteady tunnels introduce, as the name implies, a large-scale temporal variation in the test section free-stream, generally with the intent of preserving "flow quality", which is to say, to produce the desired unsteadiness without concomitant rise in broadband turbulent intensity. This is generally done by introducing a time-varying flow restriction in the tunnel, such that a given working-condition of the drive system (the tunnel fan) can be kept more or less steady, while the tunnel operating point rides up and down the pump curve, exchanging total-head for volumetric flow rate. Frequency response, or in other words fidelity between desired and attained change in test-section free stream, and lag between input and output, depends on resistance, capacitance and inductance of the tunnel circuit [7].

Typically, the time-varying flow restriction is a set of shutters, louvers or other electrically-controlled sources of blockage. Most unsteady wind tunnels have a cutoff frequency around 1Hz [6]. The louvers of the University of Dayton Low Speed Wind Tunnel (UD-LSWT) are placed just aft of the tunnel test section, with hypothesis that such placement enables higher cutoff frequency. Figure 1 depicts a conceptual idealization of an unsteady wind tunnel. There is first a transfer-function from the electrical signals for the shuttering system, to the mechanical response of

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the shutters themselves. Second, there is a transfer function from the action of the shutters to the flowfield history in the test section. Finally, there is the putative transfer function between this unsteady flow and the aerodynamic response of some test-article of interest – that is, the problem of aircraft gust response. It is philosophically appealing to regard each of the aforementioned as being at least locally linearizable, for sufficiently small perturbations about some fixed-point, and to chain the three in series. Each stage can be idealized as a spring-mass-damper.If this approach is approximately correct, one can obtain predictions of amplitude and phase of response to a sinusoidal input.

This paper considers the middle of the aforementioned systems: response of the test section free-stream, to shutter actuation. We also consider the extent to which is may be possible to introduce not just a spatially-uniform, temporally-varying streamwise gust, but a vortical gust. This would be a traveling vortex, without the undesirable attributes of a viscous wake entailed by an upstream device that normally produces such a vortex. Such a vortex, or chain of vortices, is a natural property of an open-jet wind tunnel, where the core flow is bounded by a shear layer, and ultimately the undisturbed flow in the chamber around the test section. The vortical gust is perhaps of particular relevance to ground-test simulation of atmospheric boundary-layer flow-structures encountered by small, low-altitude aircraft. [3]

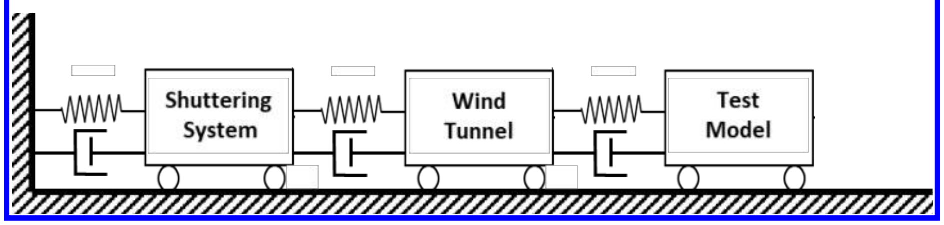

Figure 1. Abstraction of Unsteady Wind Tunnel as Spring-Mass-Damper system

2. Experimental Setup

A. Low Speed Wind Tunnel and Shuttering System

The UD-LSWT is an Eiffel-type tunnel with conventional single-stage axial fan at the downstream terminus of the diffuser. It has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections (Figure 2). One such test-section is a closed rectangular chamber or plenum circumscribing the test-area, allowing for an open-jet between the discharge from the contraction-section and the inlet of the diffuser. A photo of the UD-LSWT open jet configuration is shown in Figure 2. Stable free-stream speed in the open jet is from 6.7m/s (20 ft/s) to 40m/s (140 ft/s) at Ti% below 0.1%, as measured by hot-wire anemometer. The length of the test section in the open jet configuration is 182cm (72"), feeding into a 137cm x 137cm (44" x 44") diffuser inlet. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an Omega differential pressure transducer (Range: 0 – 6.9 kPa).

The tunnel houses an Aerotech Shuttering System just downstream of the diffuser inlet. It can vary the freestream at up to 5 Hz, and over 50% velocity amplitude. The system consists of a set of primary counter-rotating flat-plate vanes mounted in a plane normal to the flow, and a pair of side louver vane sets mounted in the sides of the diffuser. The primary louvers operate over a range of 41° from full open flow to closed flow (Figure 3). The secondary vanes are intended to minimize pumping-transients at the tunnel fan, when the primary vanes are closing. Various shutter actuation profiles are possible: sinusoidal, triangular and others. In particular, non-sinusoidal (but still smooth) shutter motion-histories are possible, to produce close approximations to sinusoidal free-stream speed variation in the test section.

B. Data Acquisition

A Dantec P5516 Probe CTA hot-wire was placed along the centerline of the test section, closer to the inlet (Figure 4). The hotwire was connected to a Dantec Mini-CTA box, and in turn to a National Instruments USB-6341 BNC. Data was sampled at 5Khz, and recorded using MATLAB's Analogue Input Recorder. Figure 4 gives typical mean hot wire results for traversing the test section in the lab-frame vertical direction, thust producing a velocity profile. The height of the "top hat" portion of this profile is taken to eb the effective test section height.

Figure 2. UD-LSWT Open-Jet Configuration

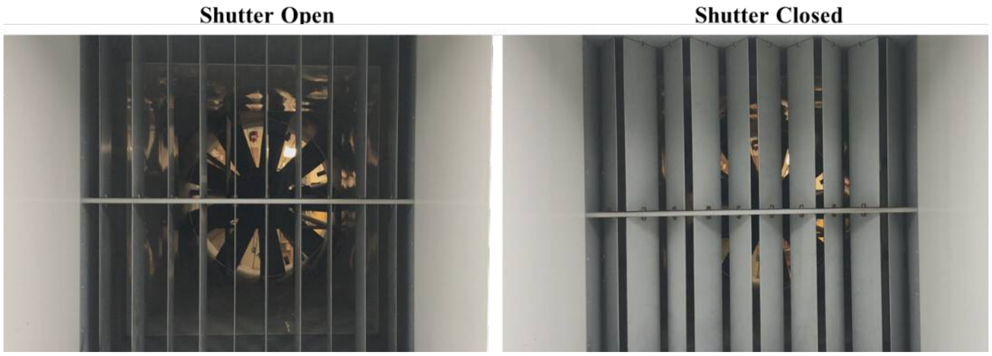

Figure 3. Shutters (louvers) downstream of the test section.

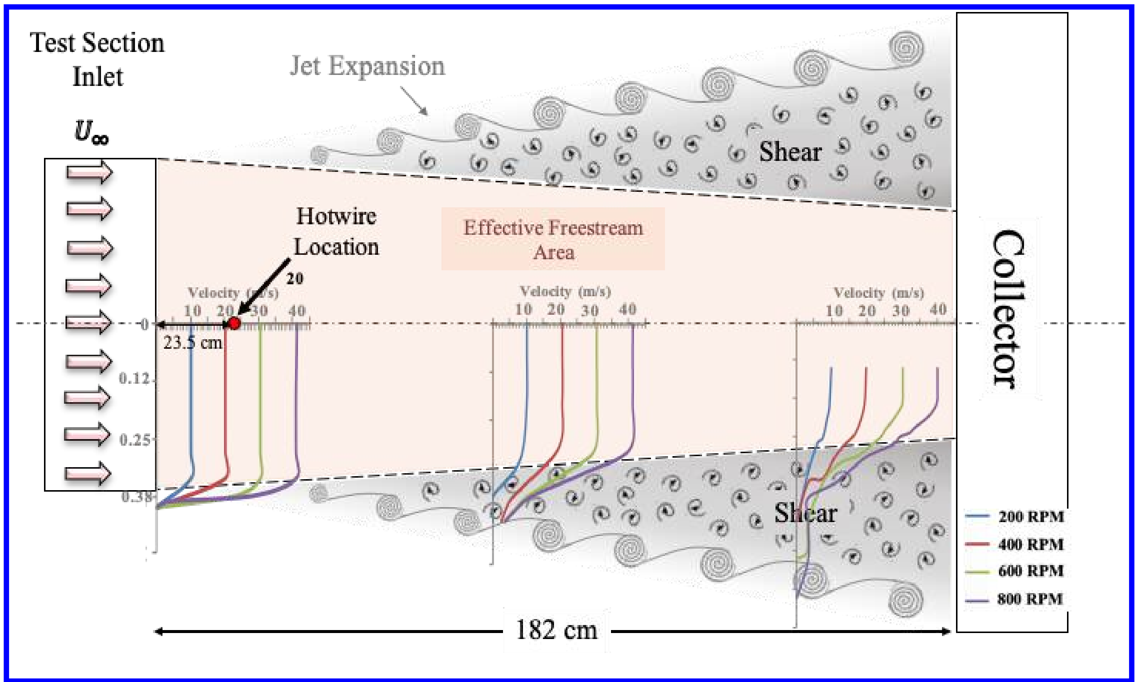

Figure 4. Location of hotwire in the UD – LSWT, and typical velocity profiles in absence of shutter actuation.

The calibration of the hot-wire was done with the test section velocity measured by a TSI 5825 Micro-anemometer at a velocity range of 2 to 30 m/s. The mean velocity was subtracted from the instantaneous velocity, yielding the fluctuating velocity. The standard deviation in the velocity data without any actuation is around 0.04 m/s.

Once the hot-wire was calibrated, a series of steady and unsteady louver actuation cases were run to characterize the wind tunnel response at the fan RPM of 150, 200, 250, 300 and 350. In the steady case, the louvers were stepped from 0^o to 41.3^o (max deflection angle) and back to 0^o in increments of 4^o. The louvers were given sinusoidal actuation input with frequencies ranging from 0.1 Hz to 4 Hz in increments of 0.2 Hz. In each case, the data was recorded for 20 seconds.

3. Results

A. Steady State Response

We next consider the effect on circuit blockage, and thus free-stream speed, with the shutters in various partially-closed positions. Test section mean flow speed, for various fan RPMs, is shown in Figure 5 as a function of louver angle. An angle of 0° is fully-open, and 45° would be fully closed. 41.3° is the maximum realizable closure-angle. As stands to reason, flow speed vs. blockage is nonlinear. The differences in the velocity between the RPM cases diminish as the shutter angle increases. The reduction in freestream velocity from the fully-open louver configuration to the maximum louver closure angle is around 72%, and is independent of tunnel-fan RPM.

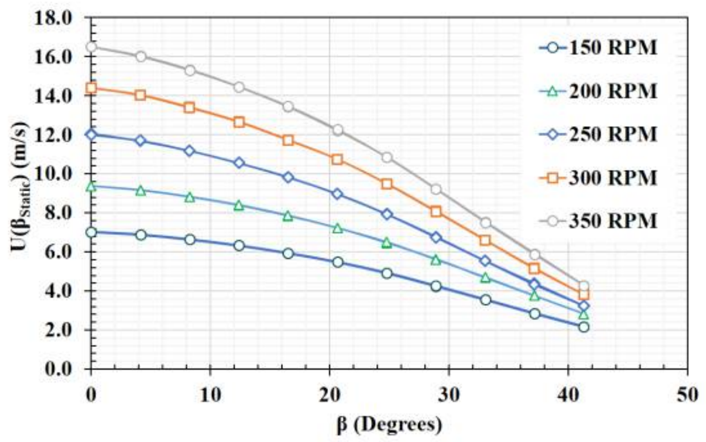

Figure 5. Test section mean flow speed vs. louver-angle, for a range of tunnel fan RPMs.

Assuming a sinusoidal motion of the louvers, the corresponding RMS of the variation would be 25.2°. The wind tunnel velocity (U_δ) at each RPM setting for shutter RMS angle of 25.2° is shown in Figure 6, with clearly linear trend.

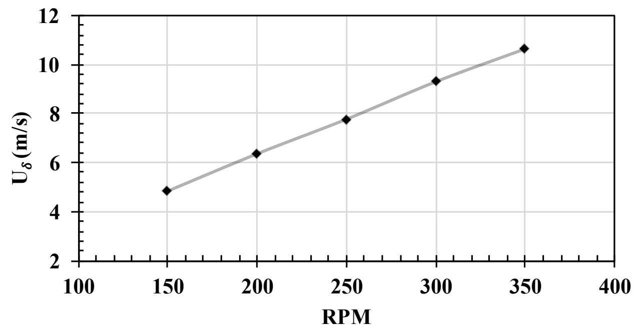

Figure 6 Test section velocity at 25.2° (RMS of the sinusoidal louver actuation)

The variation of the test section velocity in Figure 5 invites an alternative plotting, of pressure coefficient vs. discharge coefficient. The coefficient of pressure is defined as

$$C_{Pressure} = 1 - \left(\frac{U_{\beta_{Static}}}{U_{(\beta_{Static}=0)}}\right)^2 \tag{1}$$

where $U_{\beta_{Static}}$ is the mean velocity in the test section at louver angle β and $U_{(\beta_{Static}=0)}$ is the mean velocity in the test section when $\beta = 0$, which is to say, when the shutters are fully open. The volumetric flow rate, or discharge coefficient, is given by,

$$C_Q = \frac{A_{Exit}\, U_{\beta_{Static}}}{\Omega D_{Fan}^2} \tag{2}$$

where A_{Exit} is the exit area of the collector which is a function of β, Ω is the fan RPM, and D_{Fan} is the diameter of the wind tunnel fan. The trends in the pressure variation with volumetric flow rate independent of tunnel fan RPM , as seen in Figure 7.

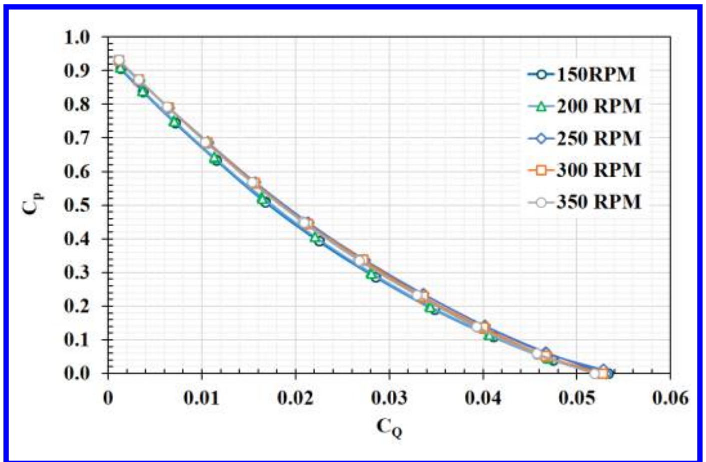

Figure 7. Pressure coefficient vs. discharge coefficient for various fan RPMs.

B. Dynamic Response

The shuttering system was given sinusoidal input from 0.1 Hz to 4 Hz, in increments of 0.2 Hz, at various tunnel RPMs. The time trace of the normalized velocity hotwire data at 150 RPM for selected frequencies is shown in **Error! Reference source not found.**. The data was taken in the centerline of the test section as indicated in Figure 4. Velocity variation is largest for the lowest shutter frequencies, with 0.1 Hz as a particularly obvious case in **Error! Reference source not found.**. As expected, the response is approximately sinusoidal. Indeed, an FFT plot shows a dominant spike at the target frequency, with some harmonics and “noise”.

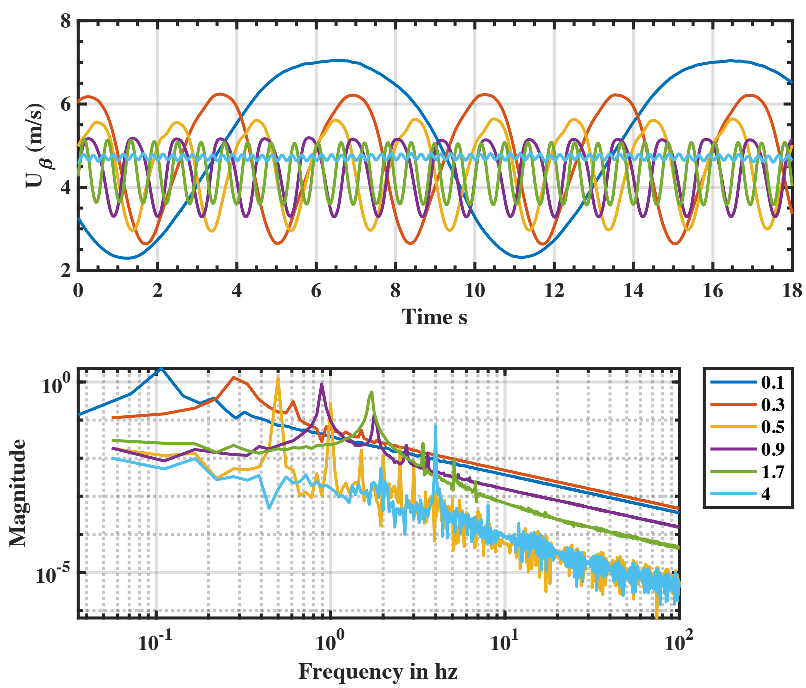

Figure 8 Time trace of the velocity in the centerline of the open jet test section at 150 RPM of the tunnel fan. B) FFT of the velocity time trace indicating the dominant frequencies in the signal.

Given the non-linear relationship between the test section velocity and the static louver angle, a pseudo-steady actuation of the louver would result in a non-linear response. Hence, a sinusoidal input to the louver would result in a non-sinusoidal response of the wind tunnel A sinusoidal input to the louver can be modeled by

$$\beta(t) = \frac{\pi}{4} + \frac{\pi}{4}\sin\left(2\pi f t - \frac{\pi}{2}\right) \tag{3}$$

The factor of $\pi/4$ in Equation 3 represents an ideal scenario where louvers actuate from 0° to 45°. In reality, the maximum actuation is restricted to an angle of 41.3° as discussed in the previous section. Here we assume a pseudo-static shutter frequency f that the velocity variation in the test section due to shutter actuation $U(\beta)$ can be predicted by the relationship between the shutter angle β and the test section velocity $U_{\beta_{Static}}$ shown in Figure 5. A generic second order polynomial regression for the $U_{\beta_{Static}}$ variation with β is

$$U_{\beta_{Pseudo-Static}} = C_1\beta(t)^2 + C_2\beta(t) + C_3 \tag{4}$$

where C_1, C_2 and C_3 are polynomial coefficients. Any higher order polynomial regression greater than 2 would result in a negligible higher order term coefficients. Therefore, a second order polynomial regression would suffice to model the behavior shown in Figure 5 with 99% accuracy. Inserting Equation 4 in Equation 3 and simplifying,

$$U_{\beta_{Pseudo-Static}} = \frac{C_1\pi^2}{4}\sin^4(\pi f t) + C_2\frac{\pi}{2}\sin^2(\pi f t) + C_3 \tag{5}$$

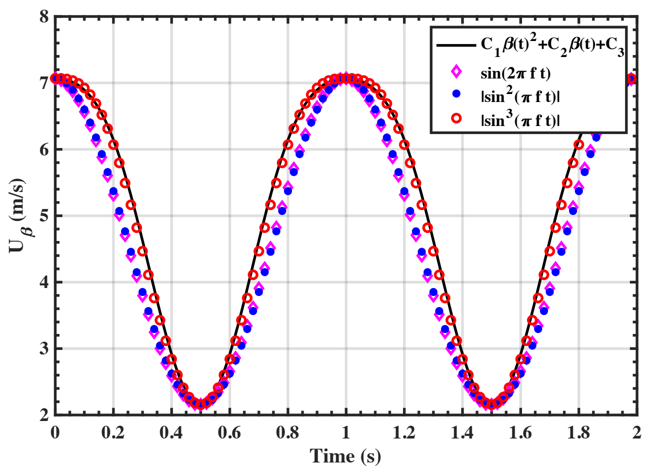

Figure 9 Prediction of instantaneous velocity in the test section at pseudo-static actuation of shutters at 150 RPM

The RHS of Equation 5 tells a compelling story of why the velocity variation in the wind tunnel test section cannot be a pure sinusoidal wave for a sinusoidal shutter input. Even though Equation 5 was derived from a pseudo-static assumption, the result will hold for a reasonably higher frequency cases since the relationship between pressure and velocity is mirrored by Equation 4. Equation 5 indicates that a pure sinusoidal input would result in a velocity profile which varies as a function of $\sin^n(t)$. Such behavior would explain the presence of strong harmonics frequencies seen in Figure 8. The instantaneous velocity variation in the open jet test section was predicted using Equation 5 at a tunnel RPM of 150 RPM. The polynomial coefficients (C_1, C_2, C_3) were found from Figure 5. Different sine wave models $f(\sin(2\pi f t))$, $f(|\sin^2(\pi f t)|)$, and $f(|\sin^3(\pi f t)|)$ were also compared with the predictions from Equation 5. It can be readily observed that the $f(|\sin^3(\pi f t)|)$ model agreed extremely well with the velocity variation predicted by Equation 5 when compared to the other two models.

Given the $f(|\sin^3(\pi f t)|)$ variation of the test section velocity predicted by Equation 5, the hotwire velocimetry data shown in Figure 8 was matched with a sine wave model raised to power of $n = 1, 2$ and 3 in an endeavor to determine the behavior of the instantaneous velocity variation in the centerline of the open jet test section for a sinusoidal input of different frequencies as shown in Equation 3 to the shutters. The matching of the experimental data with the three $\sin^n(t)$ models was performed through an optimization algorithm which determines the maximum least square fit for the three models with the experiment. The results from this analysis is shown in Figure 10 for the

150 RPM case. The calculated R^2values for each frequency case is also shown in Figure 10 for the three models. It is intriguing to observe that the behavior of the instantaneous velocity in the test section is best represented by the cubic sinusoidal model $f(|\sin^3(\pi ft)|)$ until the actuation frequency of 1 Hz as corroborated by the higher R^2values. This same trend is found in Figure 9. For frequencies greater than 1 Hz, the instantaneous velocity in the test section is better represented by the sine square model $f(|\sin^2(\pi ft)|)$as evidence by the higher R^2 values in the frequency range between 1 and 3.5. At frequencies greater than 3.5 Hz, both the pure sinusoidal model and $f(|\sin^2(\pi ft)|)$model agreed well with experimental data. At higher frequency cases (> 3.5 Hz), the non-uniform amplitude oscillation resulted in a poor match with all three models and the corresponding drop in R^2 values.

Now, we turn our attention to the mean and the RMS magnitude of the instantaneous velocity. The variation of the mean velocity with increase in shuttering frequency is shown in Figure 11 for the 150 RPM case. It can readily be seen that the mean velocity is independent of the actuation frequency with variations less than 3%. This trend is also indicated in the pseudo-static velocity model shown in Equation 3 where the mean $\overline{U_{\beta_{Pseudo-Static}}}$ would be independent of the shutter actuation frequency f. However, the amplitude of the instantaneous velocity in response to the shuttering system frequency will change as it is dependent on the distance between the shutters and the measurement location, mass flow rate and the speed of sound. The RMS of the instantaneous velocity $RMS(U_\beta)$ is represented as a shaded region in Figure 10. As the actuation frequency in increased, the $RMS(U_\beta)$ decreases while the mean stays constant. It is not surprising that a high frequency oscillation of the shutter system results in a low oscillation amplitude in the wind tunnel. It was reported in [6] that the wind tunnel acts as a low pass filter diminishing the response to high frequency oscillations of the shutter.

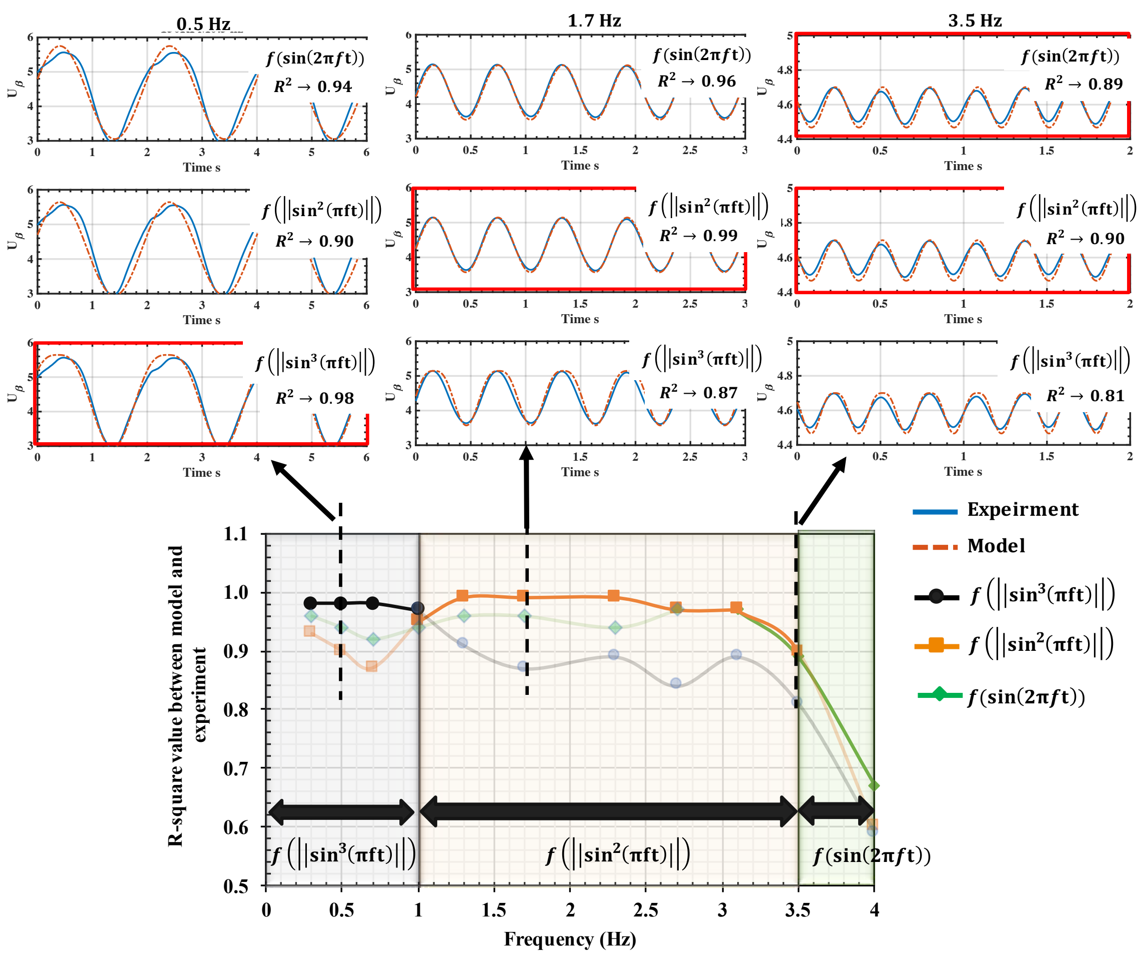

Figure 10 Matching of experimental data in the test section with sine wave models.

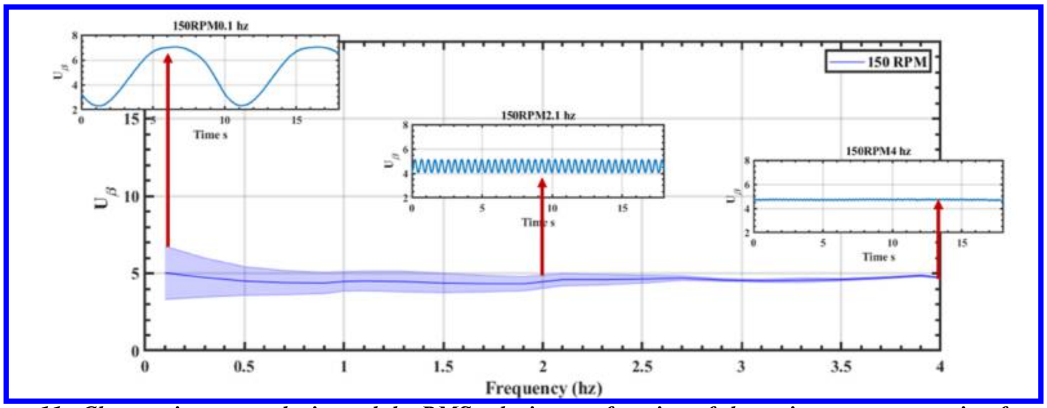

Figure 11. Changes in mean velocity and the RMS velocity as a function of shuttering system actuation frequency.

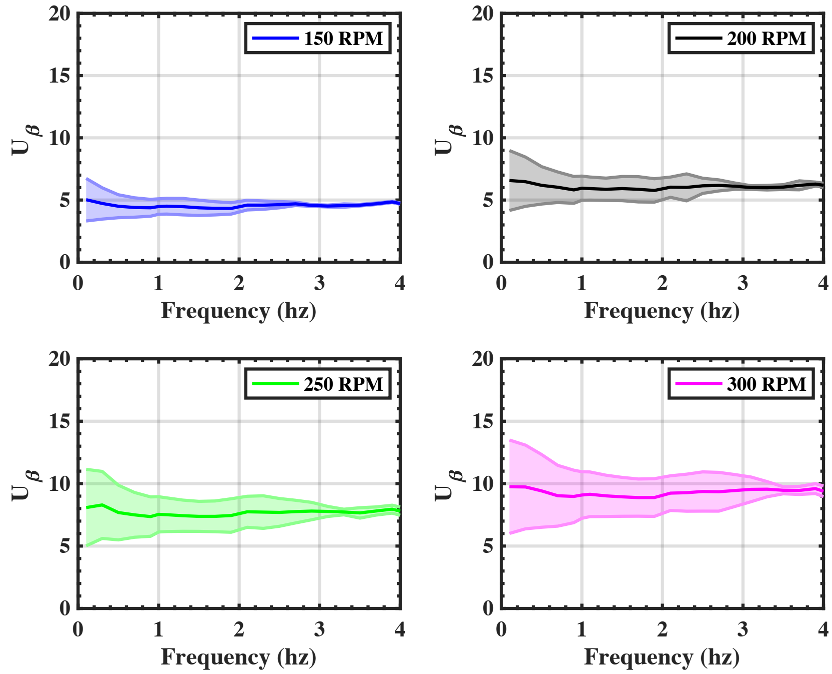

Figure 12. Mean test section velocity vs. shutter frequency. The shaded regions indicate the RMS of the velocity fluctuations.

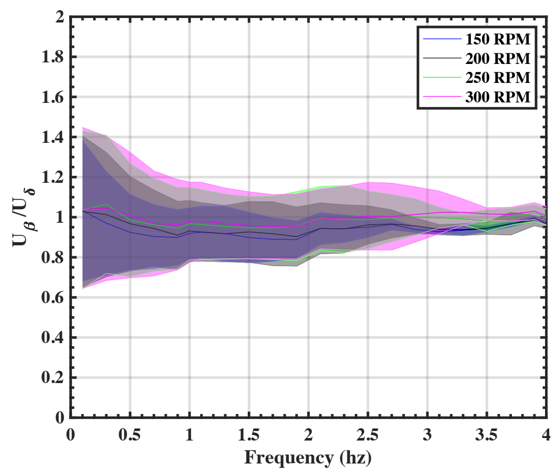

Figure 13 Normalized mean and RMS velocity variation with shutter actuation frequency

A similar representation of the mean velocity and the corresponding $RMS(U_\beta)$ as a function shutter actuation frequency is shown in Figure 12 for four different RPM cases. It can be seen that the magnitude of the mean velocity increases with the increase in the tunnel fan RPM. However, the variations in $RMS(U_\beta)$ shows similar trend for all RPM cases. The magnitude of $RMS(U_\beta)$ increase with increase in the tunnel RPM, meaning higher amplitudes in the instantaneous velocity can be achieved at higher tunnel RPMs. On an average, the ratio of the $RMS(U_\beta)$ to the mean of the instantaneous velocity $(\overline{U_\beta})$ decrease from 36% to 2% between 0.1 Hz and 4 Hz respectively.

The individual $\overline{U_\beta}$ and $RMS(U_\beta)$ for each RPM case is normalized with the tunnel velocity at $RMS(\beta)$, (U_δ) shown in Figure 6. Upon normalization, the $\overline{U_\beta}$ and $RMS(U_\beta)$ for each RPM profiles collapse onto each other as shown in Figure 13. This collapse of the velocities indicate that the mean velocity can be predicted by the pseudo-static model shown in Equation 3 with the corresponding polynomial coefficients for each RPM cases shown in Figure 5.

While the collapse of mean velocity $\overline{U_\beta}$ upon normalization is encouraging, the RMS of the velocity oscillations showed significant differences between the different RPM cases. A better visualization of the $RMS(U_\beta)$ is achieved by plotting it against the ratio of shutter frequency and tunnel RPM as shown in Figure 14. Through this normalization, it can directly be seen that the way the normalized $RMS(U_\beta)$ changes with respect to the shutter frequency is independent of the tunnel RPM until the frequency/RPM ratio of 0.005. A divergence from this trend can be seen after the frequency/RPM ratio of 0.005. Different zones of linear slopes can be gleaned as a function of frequencies in Figure 14. A negative slope of the RMS is observed in all the RPM cases until the f/Ω value of 0.005 (Zone 1). After this, a slope of zero is seen in between the f/Ω range of 0.005 and 0.01 (Zone 2) indicating the amplitude of the instantaneous velocity remains constant. This zero slope region is followed by another negative slope region (Zone 3) with increase in f/Ω. The occurrence of the different zones in the RMS can be correlated with the different behavior of the velocity variations observed in Figure 10. The Zone 1 in Figure 14 correspond to the frequencies in which the velocity variation follows the $f(|\sin^3(\pi ft)|)$ model. The Zone 2 and Zone 3 correspond to the frequencies in which the velocity variation follows the $f(|\sin^2(\pi ft)|)$ model. These variations could indicate that the natural frequency of the tunnel-shuttering system occurs at 1Hz.

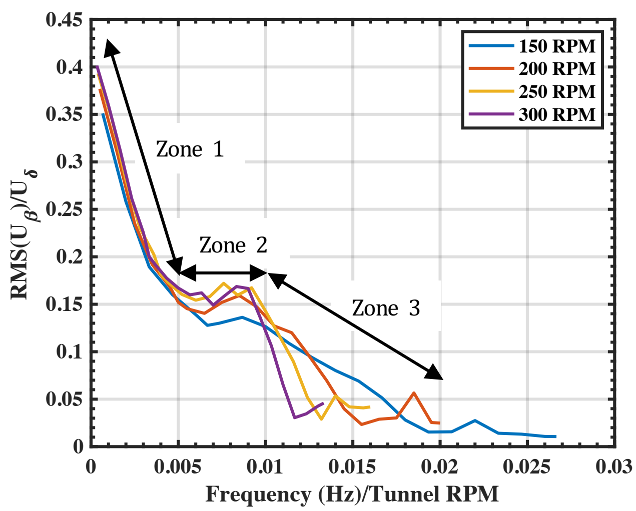

Figure 14 Variation of the normalized RMS of velocity oscillations with tunnel RPM normalized shutter frequency

C. Flow Visualization

Flow visualization with conventional smoke, assisted by a Nd:YAG continuous laser light sheet, is show in Figure 15. At 150 rpm tunnel-fanspeed, the shear layer evinces formation and downstream convection of one apparent vortex along the length from inlet-exit to collector-entrance, for shutters operating at 1 Hz. At 2 Hz, a second vortex is seen; that is to say, the frequency of vortex-eruption is such, that over the convective time covered by the relevant flow (evidently, the shear layer) at these conditions, there are two vortices in Figure 15, left column, second row from the top. At 3 Hz the inter-vortex spacing shortens, and at 4 Hz, three vortices can be observed. Doubling the tunnel reference speed to 300 fan-rpm is seen to approximatley halve the vortex formation speed. That is, at 1 Hz or 2 Hz shutter frequency, there is only one vortex visible. At 3 Hz there is speculatively onset of a condition where a new vortex forms, just as the "old" vortex is about to enter the collector. By 4Hz there are unmistably two vortices. The qualitative conclusion is that the ratio of shutter frequency to refernece tunnel speed, controls the number of visible vortices resident in the test section.

Dashd lines in Figure 15 purport a linear growth of the shear layer. This can not be quantified with snapshots of flow visualization alone, and verification awaits a more quantitative measurement. However, qualitatively, the upstream half of the test section evinces a more or less linear "wedge angle" of the shear layer, while the more downsream-half is less clear, as it is dominated by the dynamics of a vortex that has by then grown to a size of perhaps half of the test section cross-sectional height.

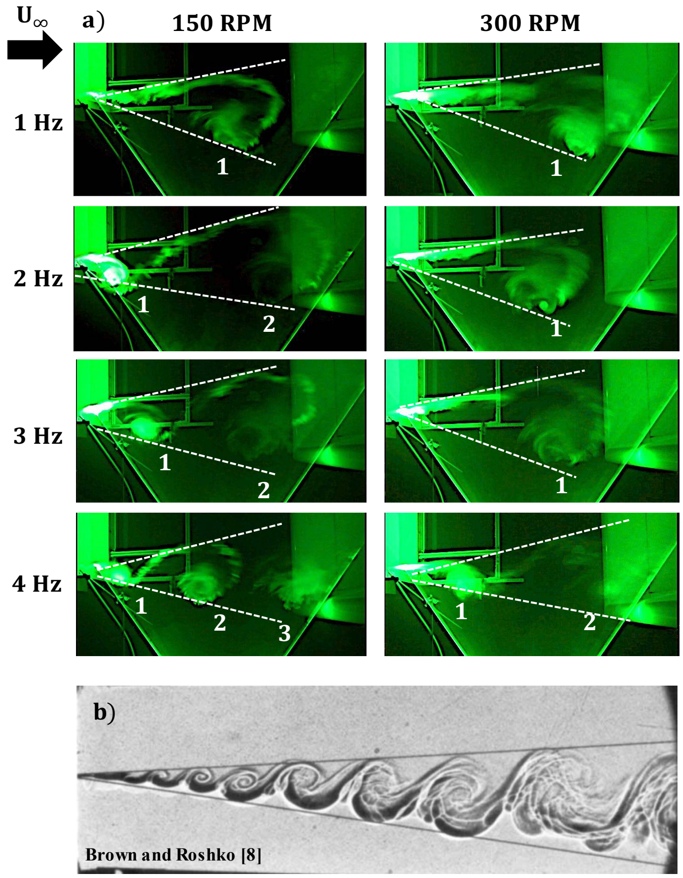

Figure 15 a) Laser light sheet flow visualization of vortex shedding. Left column: tunnel fan at 150 rpm. Right column: tunnel fan at 300 rpm.b) Shadowgraph of vortex train in mixing layer [8].

The vortex train in Figure 15a begs comparison with the classic Brown and Roshko shear layer coherent-structures [7]. For all the cases in Figure 15a, the wedge angle of the shear layer bounding the vortex train was ~20°, as compared with the ~12° wedge angle in the mixing layers in [7]. In the wind tunnel, the shear layer wedge opens more towards the tunnel floor or the ceiling, rather than into the core flow. However, at least qualitatively, a portion of the vortex enters the core flow, and it is this portion that is conjectured to bear utility in subjective test-articles of aerodynamic interest, to a vortical gust disturbance.

D. Vortex Convective Velocity

The velocity of the vortices seen in Figure 15 was found by tracking the location of the vortex center in the flow visualization video. Results for two different tunnel RPMs and two different shutter actuation frequencies are shown in Figure 16a. The x-axis of Figure 15 is the normalized vortex travel time from the inlet (left of Figure 15) to the collector (right of Figure 15). As expected, the vortex velocity increased for increase in tunnel RPM. However, for a given tunnel RPM, the vortex velocity is seen to be independent of the shutter actuation frequency. This independence of the mean vortex velocity with shutter frequency is consistent with the independence of tunnel centerline mean velocity with shutter frequency shown in Figure 11. It is also observed that the vortex accelerates slightly as it convects from the test section inlet to the collector.

The mean shear layer velocity U_{Shear} is assumed to be half of the mean velocity in the tunnel centerline and is estimated by dividing U_{δ} (shown in Figure 6) by a factor of 2. When the vortex velocity is normalized by the mean shear layer velocity, the different RPM cases collapse onto each other as shown in Figure 16b. This indicates that the vortex convective velocity is similar to that of the mean shear layer velocity and is independent of the shutter frequency.

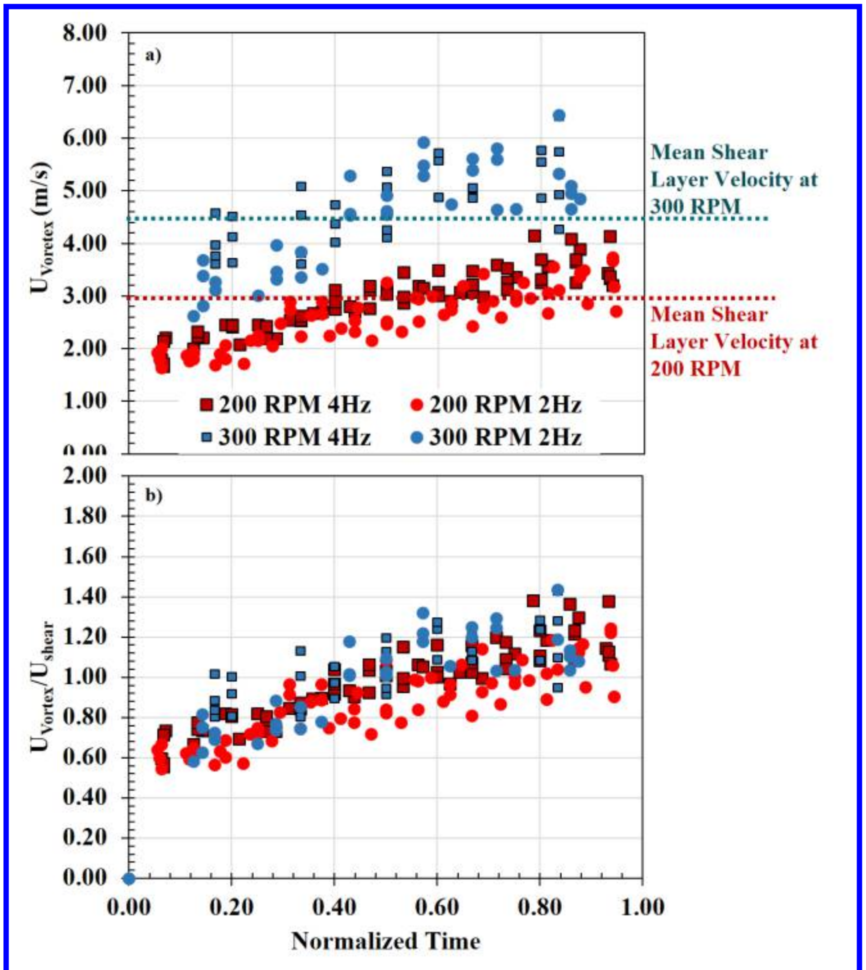

Figure 16 a) Variation of vortex velocity from inlet to collector in the test section at different RPM and shutter frequency. b) Variation of vortex velocity normalized by mean shear layer velocity.

4. Conclusions

Test section centerline free-stream speed was measured for a range of cases where a shuttering system downstream of the test section was actuated. Hot-wire data shows that the instantaneous velocity varies with as a function of $\sin^{n}(t)$ for a sinusoidal shutter input. The mean velocity in the centerline of the test section was determined to be independent of the shutter frequency, and follows the test-section velocity result for static placement of the shutters at an angle equal to that of the RMS value of the dynamic case. The amplitude/RMS of the instantaneous velocity showed strong correlations with the ratio of shutter frequency to tunnel RPM. Flow visualization by smoke-injection indicated the presence of a vortex train in the shear layer bounding the test-section core flow. The vortex velocity was determined to be similar to that of the mean shear layer velocity in the test section, and is independent of the shutter frequency. The portion of the vortex-train that enters the core flow, is conjectured to be useful for producing a vortical gust, for studies of flight-article unsteady aerodynamics.

5. References

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